

Vidhyadhari Bandaru

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EDUCATION

University at Buffalo, SUNY Master of Science in Computer Science and Engineering Relevant Coursework: Advanced Machine Learning, Natural Language Processing, Distributed Systems, Operating Systems, Computer Networks	Expected Dec 2026 GPA: 4.0 / 4.0
National Institute of Technology Rourkela, India Bachelor of Technology in Engineering Relevant Coursework: Data Structures, Artificial Intelligence, Database Systems, C Programming	Jun 2022 GPA: 3.8 / 4.0

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, C
Backend & Services: Spring Boot, FastAPI, Node.js, REST APIs, Microservices, API Design, Orchestration Workflows, Hibernate/JPA
AI & Services: LangGraph, OpenAI APIs, LLMs, RAG, Embeddings, Prompt Engineering, AST Parsing, Agentic AI
Data & Messaging: Apache Kafka, Redis, PostgreSQL, MySQL, DynamoDB, Query Optimization, Index Design, Data Modeling
Cloud & DevOps: AWS (EKS, S3, SQS, CloudWatch), Azure Container Apps, Docker, Kubernetes, Terraform, Jenkins, Git, CI/CD
Testing & OE: Unit Testing, Integration Testing, Operational Excellence (OE), Log Analysis, Debugging, Code Reviews, Agile

PROFESSIONAL EXPERIENCE

Software Engineer Capgemini Technology Services (Client: AXA Insurance) - Optimized core production Java API response times from 3.5s to 25ms by diagnosing JVM runtime bottlenecks using Java Flight Recorder, refactoring Hibernate/JPA access paths, and implementing composite PostgreSQL indexes. - Architected a resilient event-driven pipeline using multi-threaded Kafka consumers, exponential backoff retry topics, and Dead Letter Queues (DLQ), ensuring zero data loss and 99.8% event recovery during traffic spikes. - Shielded backend microservices from thread pool exhaustion by engineering a Redis-backed distributed rate limiter using Bucket4j and atomic Token Bucket validation across service instances. - Sustained 99.95% platform availability and Operational Excellence (OE) by containerizing Java microservices and managing deployment workflows using AWS EKS, Kubernetes, and Terraform. - Elevated automated test suites (JUnit, Integration, Regression), driving code coverage from 31% to 74% and detecting 20+ pre-release defects prior to UAT deployment in Agile sprint cycles. - Reduced deployment overhead by 40% by constructing Jenkins CI/CD pipelines featuring automated build validation, health checks, rollback configuration, and release gates.	Jan 2024 – Aug 2025
Associate Software Engineer Capgemini Technology Services (Client: Mars Inc.) - Designed and owned high-throughput Python/FastAPI data ingestion services processing transaction data across 180 countries with an end-to-end sub-60 second SLA and Redis-backed idempotency controls. - Built production REST APIs for inventory and reporting workflows, sustaining sub-200ms p99 response times and a 99.9% uptime SLA for global analytics and business teams. - Executed full SDLC workflows across Agile sprints—leading API design, integration testing, release coordination, UAT support, and production incident debugging. - Accelerated high-traffic endpoint performance by 25% by resolving N+1 query patterns through SQLAlchemy eager loading and query consolidation. - Validated 5M+ historical records prior to release by engineering Databricks PySpark data pipelines that identified data integrity anomalies and processing regressions. - Streamlined root-cause analysis and incident resolution by 30% by implementing structured JSON logging, metrics collection, and real-time alerts in AWS CloudWatch.	Sep 2022 – Dec 2023

PROJECTS

PR Blast Radius Analyzer: AI Orchestration Tool <i>Python, FastAPI, DynamoDB, LangGraph, OpenAI APIs, Azure</i> - Accelerated cross-repository code analysis from 2+ hours to under 15 seconds by engineering an AI orchestration tool with FastAPI, tree-sitter AST parsing, and LangGraph workflow routing. - Improved operational readiness and developer productivity by caching symbol dependency trees in DynamoDB and implementing prompt-injection guards for AI outputs.	May 2026
Socratic NLP Tutor: AI Workflow Engine <i>Python, FastAPI, OpenAI APIs, RAG, BM25, Vector Search</i> - Built a high-accuracy tutoring engine reducing direct-answer leakage by 100% using FastAPI, structured prompt constraints, and recursive Socratic questioning workflows. - Enhanced context retrieval precision by combining BM25 lexical search, vector embeddings, and cross-encoder reranking to ground LLM responses in factual course data.	Spring 2026
Commerce Transaction Microservices Platform <i>Java, Spring Boot, Kafka, PostgreSQL, Redis, AWS, Docker, Kubernetes</i> - Maintained 100% data consistency across 10K concurrent transactions by building a 4-service Java/Spring Boot backend using the Saga Pattern and Transactional Outbox pattern. - Prevented state corruption during node failures by implementing partition-aware Kafka event consumers and Redis-backed idempotent processing.	June 2026

LEADERSHIP & RECOGNITION

Capgemini Best Employee of the Quarter Award Recognized for delivery ownership, customer satisfaction, cross-functional communication, and improvements to production backend reliability.	Mar 2025
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